Unit: 6 Unit 8: Quadratics **Name:** ________________

Topic: Systems of Quadratic Equations **Date:** ___________

Advanced (Habanero)

Q1. Solve the system: $x^2 + y^2 = 4$, $x^2 - y^2 = -1$

(A) $(-1, -sqrt3)$

(B) $(-1, sqrt3)$

(C) $(1, -sqrt3)$

(D) $(1, sqrt3)$

(E) No solution

Q2. Solve the system: $y = x^2 - 2$, $y = -x^2 + 2$

(A) $(0, 2)$

(B) $(1, 0)$

(C) $(-1, 0)$

(D) $(0, -2)$

(E) Infinite solutions

Q3. The graphs of $y = x^2 + 2x - 3$ and $y = -x^2 - 2x + 3$ intersect at

(A) One point

(B) Two points

(C) Three points

(D) Four points

(E) No points

Q4. Solve the system: $x^2 + y^2 = 25$, $x^2 - y^2 = 7$

(A) $(2, -3)$

(B) $(2, 3)$

(C) $(-2, -3)$

(D) $(-2, 3)$

(E) $(4, 3)$

Q5. The system of equations $y = x^2 - 4$, $y = -x^2 + 4$ has

(A) One solution

(B) Two solutions

(C) Three solutions

(D) Four solutions

(E) No solutions

Q6. The graphs of $y = x^2 - 2x - 3$ and $y = x^2 + 2x - 3$ intersect at

(A) One point

(B) Two points

(C) $(-1, -4)$

(D) $(1, -4)$

(E) No points

Q7. Solve the system: $y = x^2 - 3$, $y = x^2 + 3$

(A) $(0, 0)$

(B) $(0, 3)$

(C) $(1, 0)$

(D) $(-1, 0)$

(E) No solution

Q8. The system of equations $y = x^2 + 1$, $y = -x^2 - 1$ has

(A) One solution

(B) Two solutions

(C) Three solutions

(D) Four solutions

(E) No solutions

Open Ended Questions (FRQ)

Q9. Graph the system of equations $y = x^2 + 2x - 3$, $y = x^2 - 2x - 3$. Find the intersection points.

Q10. Solve the system of equations $x^2 + y^2 = 20$, $x^2 - y^2 = -5$. Check your solution graphically.

Chef's Tips

- Emphasize the importance of graphing the equations to visualize the solutions
- Encourage students to use algebraic methods to solve the systems of equations
- Remind students to check their solutions by plugging them back into the original equations